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Unit 3 : Session 4 : SLO 1 — Shift-and-Add Unsigned Multiplication

■ 1. Vocabulary Match-Up (Remembering)

1. Multiplicand → C. The number that is being multiplied
2. Multiplier → D. The number by which you multiply
3. Partial Product → E. Result of multiplying the multiplicand by a single multiplier bit
4. Accumulator → B. Register that holds the intermediate product
5. Shift-Left → F. Moves all bits one position to the left
6. Product → A. The result of the multiplication

■ 2. Step-by-Step Description (Understanding) — correct order

- 1 → Initialize the Accumulator to all zeros.
- 2 → If the LSB of the Multiplier is 1, add the Multiplicand to the Accumulator.
- 3 → Shift the Multiplicand one position to the left.
- 4 → Shift the Multiplier one position to the right.
- (In hardware, steps 2–4 form one cycle repeated for each multiplier bit.)

■ 3. Applying the Concept (Applying): 1101 (13) × 1011 (11) by shift-and-add

- Initial: Multiplicand = 0000 1101, Multiplier = 1011, Accumulator = 0000 0000
- Iteration 1: LSB = 1 → Add: Accumulator = 0000 1101; then shift → Mcand = 0001 1010, Mplier = 0101
- Iteration 2: LSB = 1 → Add shifted multiplicand: 0000 1101 + 0001 1010 = 0010 0111 (Accumulator = 0010 0111); shift → Mcand = 0011 0100, Mplier = 0010
- Iteration 3: LSB = 0 → No add; shift → Mcand = 0110 1000, Mplier = 0001
- Iteration 4: LSB = 1 → Add shifted multiplicand: 0010 0111 + 0110 1000 = 1000 1111
- Final Binary Product: 10001111 Final Decimal Product: 143 ($13 \times 11 = 143$)

■ 4. Error Analysis (Analyzing): 101 × 110

- Incorrect step number: Step 3.
- Error: in Step 3 the accumulator holds 001010 and the shifted multiplicand is 010100; since the multiplier LSB is 1 the sum must be $001010 + 010100 = 011110$. The student wrote 011100 — a binary addition error.

■ 5. Design a Problem (Creating)

- Chosen 3-bit multiplicand: 111 (7) Chosen 3-bit multiplier: 101 (5)
- Step 1: LSB = 1 → add → Accumulator = 000111; shift → Mcand = 001110, Mplier = 010
- Step 2: LSB = 0 → no add; shift → Mcand = 011100, Mplier = 001
- Step 3: LSB = 1 → add → $000111 + 011100 = 100011$; shift → done
- Final Answer: 100011 (binary) = 35 (decimal)